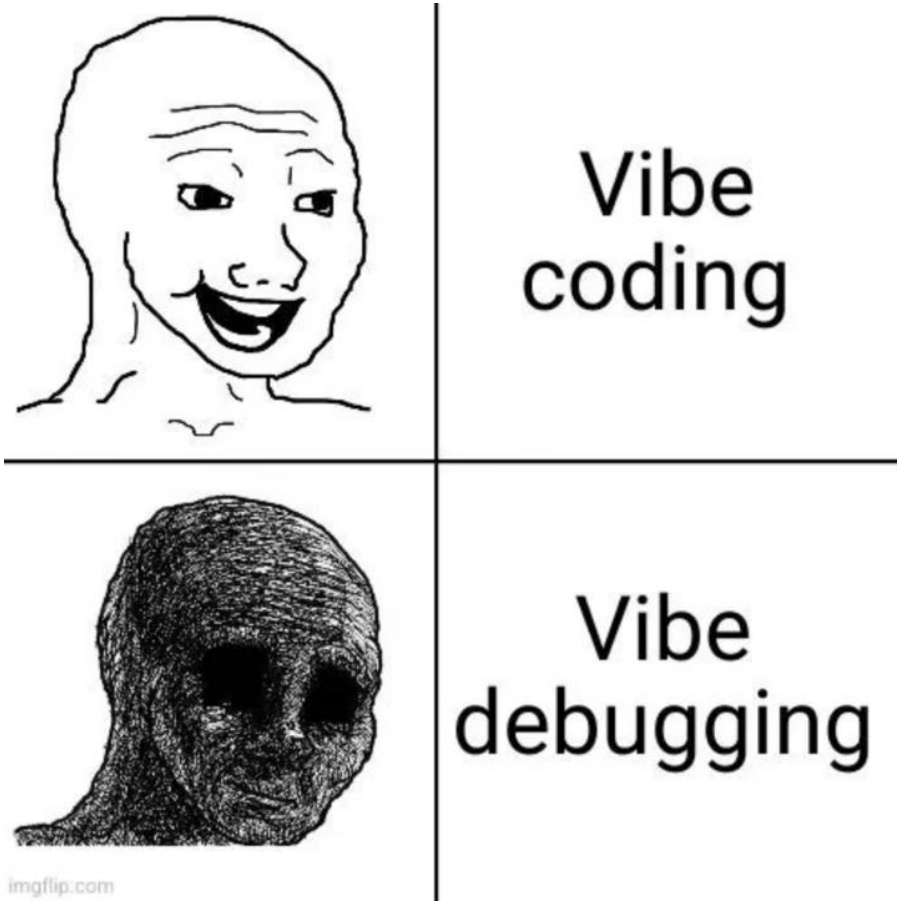
# Lecture 7 – Complex Python Syntax



# Today's Learning Outcomes

1. Be able to read and use for and while loops

2. Be able to read and use if-else statements

3. Systematic overview of the in-class practice codes for the first six lectures

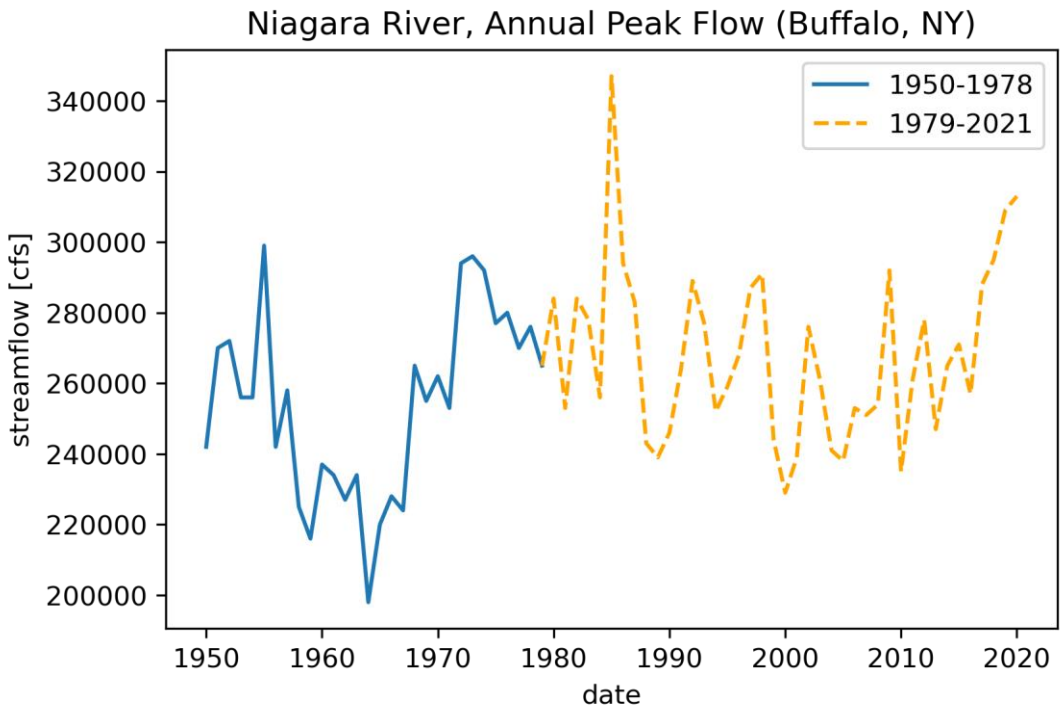
# Questions from last class: T test

## Why do we need T-test?

- Are recent temperatures warmer than historical averages?
- Did a restoration project increase stream oxygen?
- Does vegetation NDVI differ between burned and unburned areas?
- Is snowpack lower this year than the long-term mean?

**A t-test answers the question: Are two averages different beyond what random chance can explain?**

# Does Niagara River have a higher peak flow post-1980?




The figure on the left shows the timeseries of annual peak flow from 1950 till recently.

Blue line denotes prior to 1980, and yellow dashed line denotes post-1980.

What is null hypothesis?

$$\bar{Q}_{peak,prior-1980} = \bar{Q}_{peak,post-1980}$$

***What might contribute to how confident we can reject the null hypothesis?***

# *What might contribute to how confident we can reject the null hypothesis?*

1. Difference between means
   - big difference → easier to detect
2. Variability within each group
   - noisy data → harder to detect
3. Sample size
   - more data → easier to detect

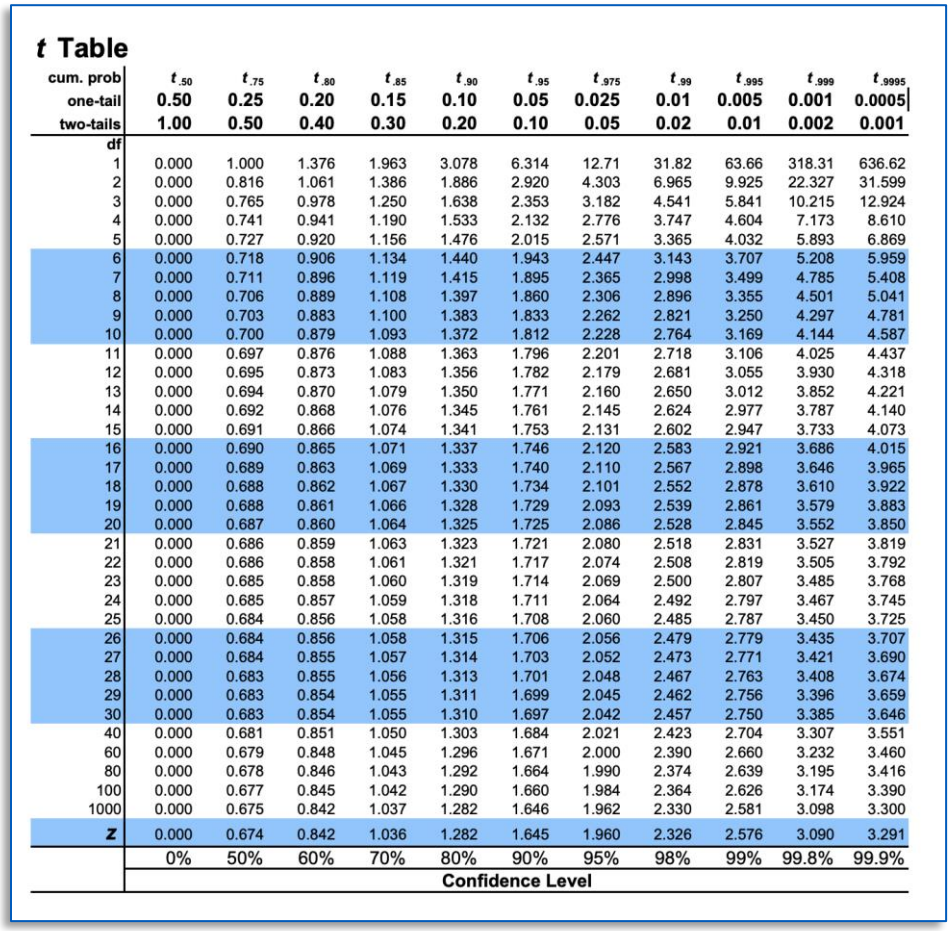
**T score equation**

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

where

$$s_p = \sqrt{\frac{(n_1 - 1)s_{X_1}^2 + (n_2 - 1)s_{X_2}^2}{n_1 + n_2 - 2}}$$

# After obtaining T score, we can use look-table to find the p-value



**t Table**

| cum. prob | $t_{.50}$ | $t_{.75}$ | $t_{.80}$ | $t_{.85}$ | $t_{.90}$ | $t_{.95}$ | $t_{.975}$ | $t_{.99}$ | $t_{.995}$ | $t_{.999}$ | $t_{.9995}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| one-tail | 0.50 | 0.25 | 0.20 | 0.15 | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 | 0.001 | 0.0005 |
| two-tails | 1.00 | 0.50 | 0.40 | 0.30 | 0.20 | 0.10 | 0.05 | 0.02 | 0.01 | 0.002 | 0.001 |
| df | | | | | | | | | | | |
| 1 | 0.000 | 1.000 | 1.376 | 1.963 | 3.078 | 6.314 | 12.71 | 31.82 | 63.66 | 318.31 | 636.62 |
| 2 | 0.000 | 0.816 | 1.061 | 1.386 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 22.327 | 31.599 |
| 3 | 0.000 | 0.765 | 0.978 | 1.250 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 10.215 | 12.924 |
| 4 | 0.000 | 0.741 | 0.941 | 1.190 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 7.173 | 8.610 |
| 5 | 0.000 | 0.727 | 0.920 | 1.156 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 5.893 | 6.869 |
| 6 | 0.000 | 0.718 | 0.906 | 1.134 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 | 5.208 | 5.959 |
| 7 | 0.000 | 0.711 | 0.896 | 1.119 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 | 4.785 | 5.408 |
| 8 | 0.000 | 0.706 | 0.889 | 1.108 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 | 4.501 | 5.041 |
| 9 | 0.000 | 0.703 | 0.883 | 1.100 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 | 4.297 | 4.781 |
| 10 | 0.000 | 0.700 | 0.879 | 1.093 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 | 4.144 | 4.587 |
| 11 | 0.000 | 0.697 | 0.876 | 1.088 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 | 4.025 | 4.437 |
| 12 | 0.000 | 0.695 | 0.873 | 1.083 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 | 3.930 | 4.318 |
| 13 | 0.000 | 0.694 | 0.870 | 1.079 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 | 3.852 | 4.221 |
| 14 | 0.000 | 0.692 | 0.868 | 1.076 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 | 3.787 | 4.140 |
| 15 | 0.000 | 0.691 | 0.866 | 1.074 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 | 3.733 | 4.073 |
| 16 | 0.000 | 0.690 | 0.865 | 1.071 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 | 3.686 | 4.015 |
| 17 | 0.000 | 0.689 | 0.863 | 1.069 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 | 3.646 | 3.965 |
| 18 | 0.000 | 0.688 | 0.862 | 1.067 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 | 3.610 | 3.922 |
| 19 | 0.000 | 0.688 | 0.861 | 1.066 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 3.579 | 3.883 |
| 20 | 0.000 | 0.687 | 0.860 | 1.064 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 3.552 | 3.850 |
| 21 | 0.000 | 0.686 | 0.859 | 1.063 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 | 3.527 | 3.819 |
| 22 | 0.000 | 0.686 | 0.858 | 1.061 | 1.321 | 1.717 | 2.074 | 2.508 | 2.819 | 3.505 | 3.792 |
| 23 | 0.000 | 0.685 | 0.858 | 1.060 | 1.319 | 1.714 | 2.069 | 2.500 | 2.807 | 3.485 | 3.768 |
| 24 | 0.000 | 0.685 | 0.857 | 1.059 | 1.318 | 1.711 | 2.064 | 2.492 | 2.797 | 3.467 | 3.745 |
| 25 | 0.000 | 0.684 | 0.856 | 1.058 | 1.316 | 1.708 | 2.060 | 2.485 | 2.787 | 3.450 | 3.725 |
| 26 | 0.000 | 0.684 | 0.856 | 1.058 | 1.315 | 1.706 | 2.056 | 2.479 | 2.779 | 3.435 | 3.707 |
| 27 | 0.000 | 0.684 | 0.855 | 1.057 | 1.314 | 1.703 | 2.052 | 2.473 | 2.771 | 3.421 | 3.690 |
| 28 | 0.000 | 0.683 | 0.855 | 1.056 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 | 3.408 | 3.674 |
| 29 | 0.000 | 0.683 | 0.854 | 1.055 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 | 3.396 | 3.659 |
| 30 | 0.000 | 0.683 | 0.854 | 1.055 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 | 3.385 | 3.646 |
| 40 | 0.000 | 0.681 | 0.851 | 1.050 | 1.303 | 1.684 | 2.021 | 2.423 | 2.704 | 3.307 | 3.551 |
| 60 | 0.000 | 0.679 | 0.848 | 1.045 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 | 3.232 | 3.460 |
| 80 | 0.000 | 0.678 | 0.846 | 1.043 | 1.292 | 1.664 | 1.990 | 2.374 | 2.639 | 3.195 | 3.416 |
| 100 | 0.000 | 0.677 | 0.845 | 1.042 | 1.290 | 1.660 | 1.984 | 2.364 | 2.626 | 3.174 | 3.390 |
| 1000 | 0.000 | 0.675 | 0.842 | 1.037 | 1.282 | 1.646 | 1.962 | 2.330 | 2.581 | 3.098 | 3.300 |
| z | 0.000 | 0.674 | 0.842 | 1.036 | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 3.090 | 3.291 |
| | 0% | 50% | 60% | 70% | 80% | 90% | 95% | 98% | 99% | 99.8% | 99.9% |
| | Confidence Level | | | | | | | | | | |

Or we have a much easier way to perform T test, which is to use the `ttest_ind` in Python!

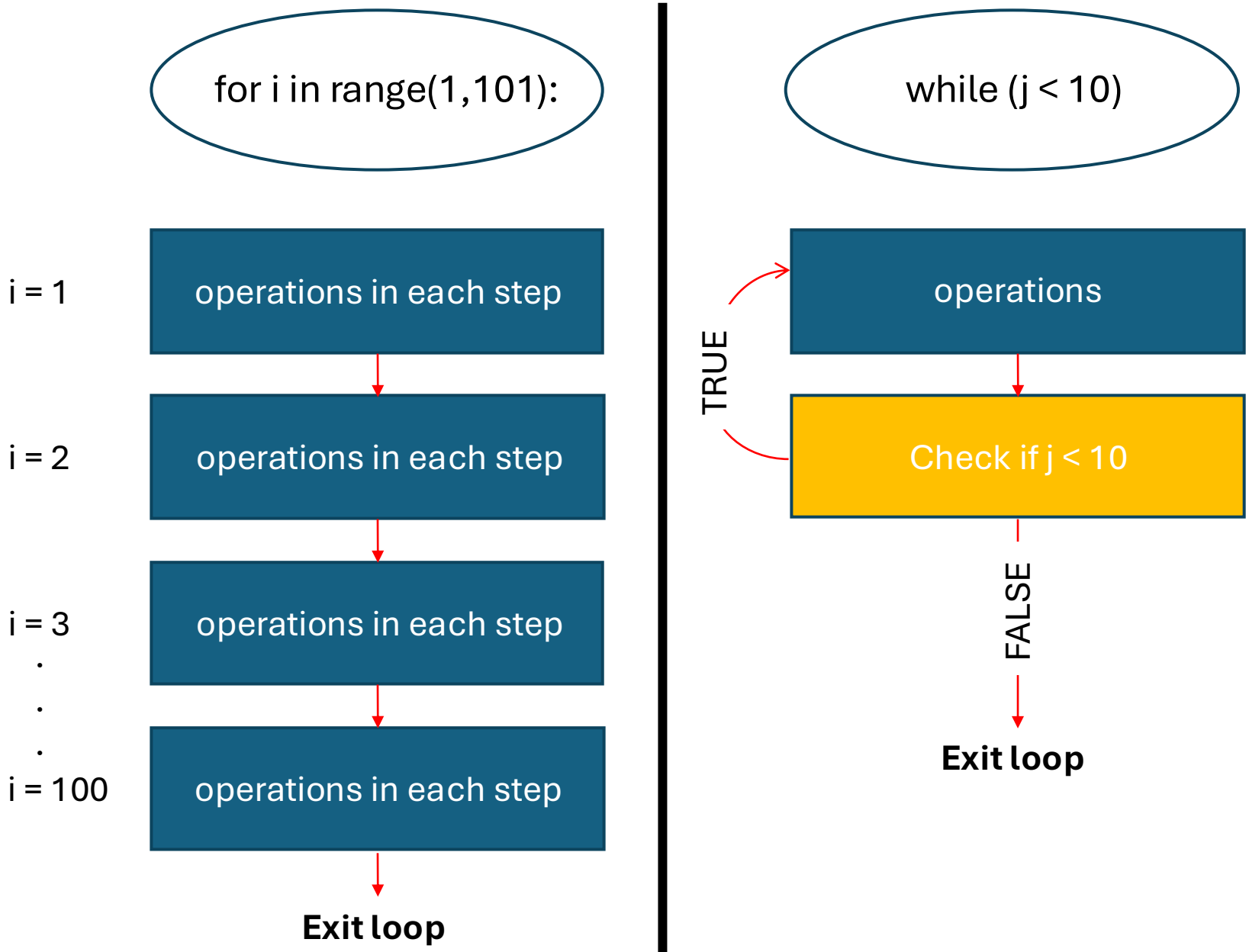
# Example code for performing T test in Python

```
from scipy.stats import ttest_ind
before = [...]
after = [...]
t, p = ttest_ind(before, after)
print(t, p)
```

# Loops in Python

- Loops are special commands which are used to simplify the process of doing the same, or slight iterations of, the same calculations many times
  - Many simple calculations is what computers excel at!

- Two main forms in Python
  - for loop; steps through a certain number of iterations
  - while loop; loop continues until some condition is met


for i in range(1,101):
i = 1
operations in each step
i = 2
operations in each step
i = 3
operations in each step
i = 100
operations in each step
Exit loop
while (j < 10)
operations
TRUE
Check if j < 10
FALSE
Exit loop

# Syntax for loops (use for-loop as an example)

**Loop through a range of numbers**

```
for i in range(1,10):
    print(i)
```

**Loop through a list**

```
ln=['Tom','Abby','Cathy']
for name in ln:
    print(name)

for i,name in enumerate(ln):
    print(f"order is {i}")
    print(f"name is {name}"
```

**Loop through two lists as pairs**

```
ln=['Tom','Abby','Cathy']
ls=[70,85,76]
for name,score in zip(ln,ls):
    print(f"The score for
{name} is {score}"
```

**Loop through Pandas DataFrame**

```
for row in df.iterrows():
    print(f"Index is {row[0]}")
    print(f"Value is
{row[1][Columnname]}")
```

# **Pandas** Data Structures

| Data Structure | Dimension |
|---|---|
| Series | 1 |
| Data Frames | 2 |

**What is the difference between series and Data Frames?**

## Data Series

```
tom       105
bob       306
nancy    3560
dan      1200
eric       50
```

## Data Framework

| | Fav_number | Fav_color |
|---|---|---|
| **tom** | 105 | red |
| **bob** | 306 | blue |
| **nancy** | 3560 | orange |
| **dan** | 1200 | pink |
| **eric** | 50 | green |

# DataFrame syntax

column

Row

| | Fav_number | Fav_color |
|---|---|---|
| **tom** | 105 | red |
| **bob** | 306 | blue |
| **nancy** | 3560 | orange |
| **dan** | 1200 | pink |
| **eric** | 50 | green |

# DataFrame syntax

Column name

index

| | Fav_number | Fav_color |
|---|---|---|
| **tom** | 105 | red |
| **bob** | 306 | blue |
| **nancy** | 3560 | orange |
| **dan** | 1200 | pink |
| **eric** | 50 | green |

# DataFrame syntax

**df.loc[index, column name]**

Column name

index

| | Fav_number | Fav_color |
|---|---|---|
| **tom** | 105 | red |
| **bob** | 306 | blue |
| **nancy** | 3560 | orange |
| **dan** | 1200 | pink |
| **eric** | 50 | green |

# DataFrame syntax

**df.loc['tom','Fav_number']**

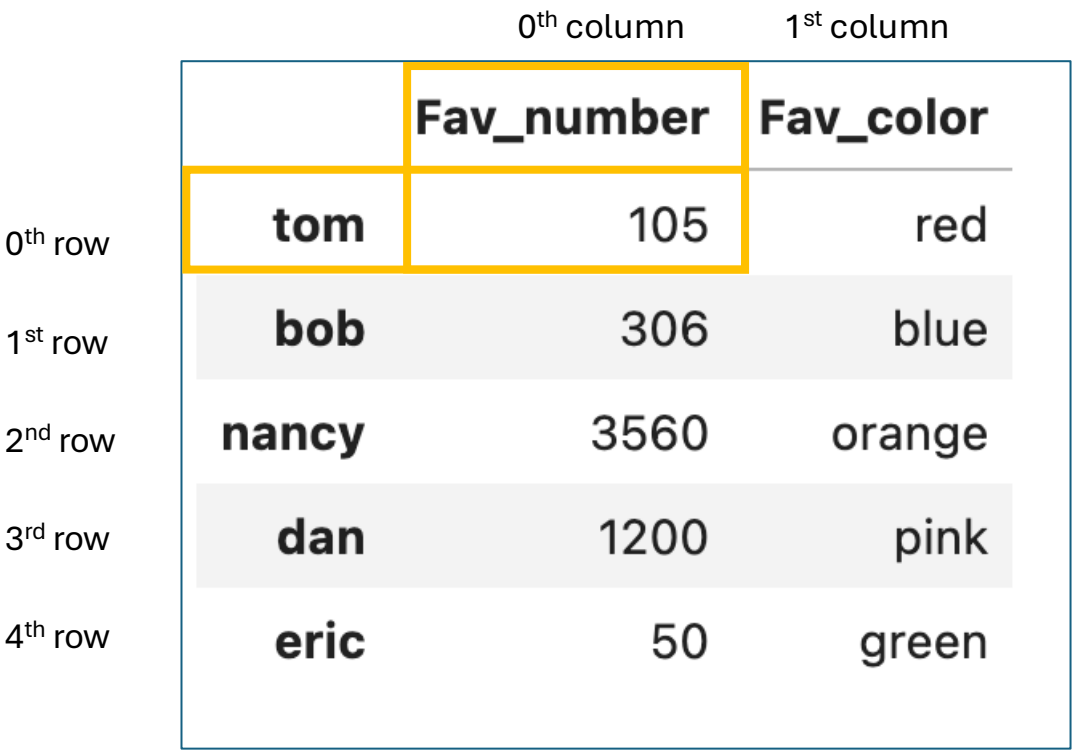
Column name

index

| | Fav_number | Fav_color |
|---|---|---|
| **tom** | 105 | red |
| **bob** | 306 | blue |
| **nancy** | 3560 | orange |
| **dan** | 1200 | pink |
| **eric** | 50 | green |

# DataFrame syntax

**df.iloc[0,0]**




| | Fav_number | Fav_color |
|---|---|---|
| **tom** | 105 | red |
| **bob** | 306 | blue |
| **nancy** | 3560 | orange |
| **dan** | 1200 | pink |
| **eric** | 50 | green |

# For-loop for DataFrame

| | Fav_number | Fav_color |
|---|---|---|
| **tom** | 105 | red |
| **bob** | 306 | blue |
| **nancy** | 3560 | orange |
| **dan** | 1200 | pink |
| **eric** | 50 | green |

**How to loop through this DataFrame and list everyone's favorite colors?**

```
>>>for row in df.iterrows():
>>>    name = row[0]
>>>    fav_color = row[1]['Fav_color']
>>>    print(f"The favorite color of {name} is {fav_color}")
```

# `if` statement

Earth scientists constantly make conditional decisions:

- Is this precipitation event likely snow or rain?
- Does this river exceed flood stage?
- Is this satellite pixel cloudy?
- Is temperature above the critical threshold for permafrost thaw?

**An if statement lets the computer make decisions like you would when classifying environmental conditions.**

# `if` statement – example 1

Flood Warning From Streamflow

- The flow threshold to predict a flood warning is 500 cfs, let's write an `if` statement to identify whether a given flow will trigger flood warning

```
>>> streamflow = 560  # cubic feet per second(cfs)
>>> flood_stage = 500

>>> if streamflow > flood_stage:
>>>     print("Flood conditions!")
>>> else:
>>>     print("Normal flow")
```

# `if` statement – example 2

Air Quality Classification

- If PM2.5 index is not exceeding **12**, the air quality is classified as **Good**; otherwise, if PM2.5 index is not exceeding **35**, **Moderate**; otherwise, if PM2.5 index is not exceeding **55**, **Unhealthy for sensitive groups**; if PM2.5 is above **55**, **Unhealthy**.

```
>>> pm25 = 68  # µg/m³

>>> if pm25 <= 12:
>>>     print("Good")
>>> elif pm25 <= 35:
>>>     print("Moderate")
>>> elif pm25 <= 55:
>>>     print("Unhealthy for sensitive groups")
>>> else:
>>>     print("Unhealthy")
```

# Today's Learning Outcomes

1. Be able to read and use for and while loops

2. Be able to read and use if-else statements

3. Systematic overview of the in-class practice codes for the first six lectures